

Polynomial Regression

Assignment 1 | BT2024260 | 8 October 2026

1. Aim of the assignment

The aim is to predict a score from the given inputs using polynomial regression. Two separate models are needed: one for the power plant settings (var1) and one for the underground location coordinates (var2).

Problem	Inputs	Score to predict
var1	Six plant settings: x1 to x6	Net Power Score
var2	Three coordinates: x1 to x3	Thermal Anomaly Score

2. Data preparation

Only the datasets for roll number BT2024260 were used. Each problem has 1,000 training rows and 1,000 test rows. There were no missing values. All input values lie between -1 and 1.

Some rows share the same inputs. These rows were kept together when splitting the data, so the same setting or location could not appear in both the fitting and validation parts. All observed responses were retained.

3. Final models

Problem	Method	Degree	Alpha	Nonzero terms
var1	Lasso	5	0.01	114
var2	Ridge	13	0.316228	559

The degree controls how complex the polynomial can be. Alpha controls regularization, which discourages overly large coefficients. Nonzero terms are the polynomial terms that remain active in the final model.

Both methods produce polynomial regression models. Ridge reduces coefficient sizes. Lasso can also remove terms by setting their coefficients to zero. Each problem uses the method with the lower cross-validation error.

Choosing the models

A fair comparison of Ridge and Lasso

4. How the comparison was done

About 20% of the input groups were set aside as a holdout: 200 rows for var1 and 198 for var2. The remaining rows were split into three validation folds using seed 42. Each candidate was fitted on two folds and checked on the third, repeating this three times. The average MSE was used to choose the model.

For both methods, degrees 1-10 were tested for var1 and degrees 1-20 for var2. Ridge used 23 alpha values from 10^{-8} to 10^3 . Lasso used 19 values from 0.001 to 1 for var1 and 25 from 0.0001 to 1 for var2. The values were logarithmically spaced. Every candidate used the same data splits.

Lasso features were centered and scaled using only the fitting fold. Ridge features were centered without variance scaling. Both used an unpenalized intercept. The Lasso calculations were checked before comparing scores.

Problem	Best Ridge CV MSE	Best Lasso CV MSE	Selected
var1	0.584824	0.332479	Lasso
var2	0.339568	0.368371	Ridge

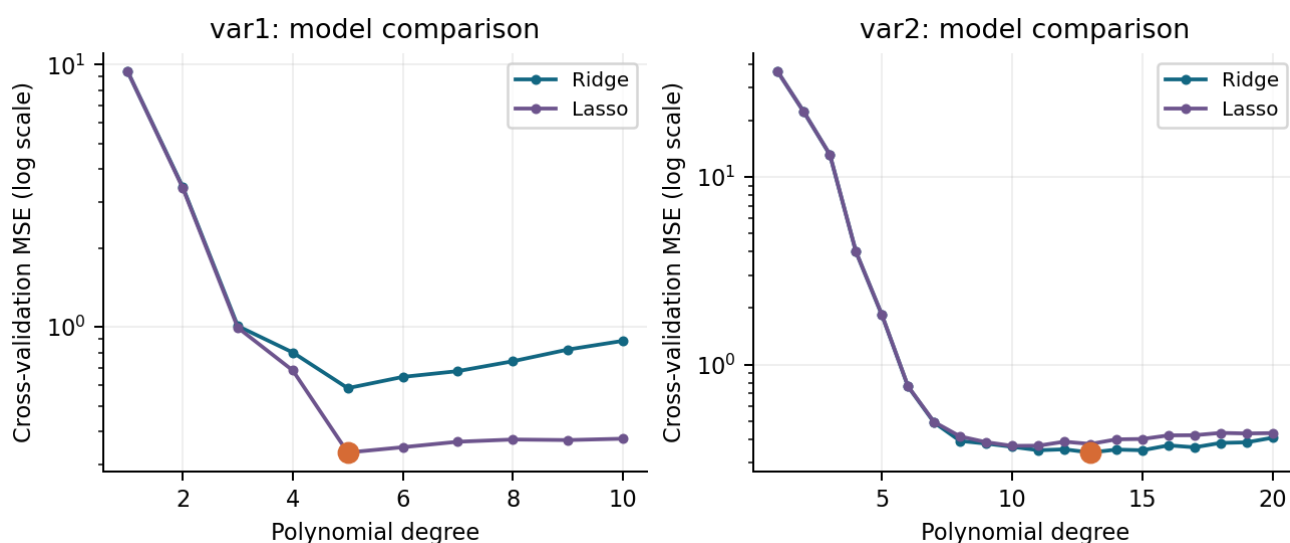


Figure 1. Lower MSE is better. Each curve shows the best alpha at each degree. The orange dot marks the selected model. The vertical scale is logarithmic.

The existing holdout had already been viewed in the earlier Ridge analysis. It was reused for the checks below, but not to choose the method, degree, or alpha.

Checking the results

Scores and plots on the holdout rows

MSE: the average squared prediction error; lower is better. **R2:** how much target variation the predictions explain; values closer to 1 are better.

Problem	Holdout MSE	Holdout R2	Mean-only MSE
var1	0.371759	0.971201	12.987200
var2	0.252532	0.994440	46.160727

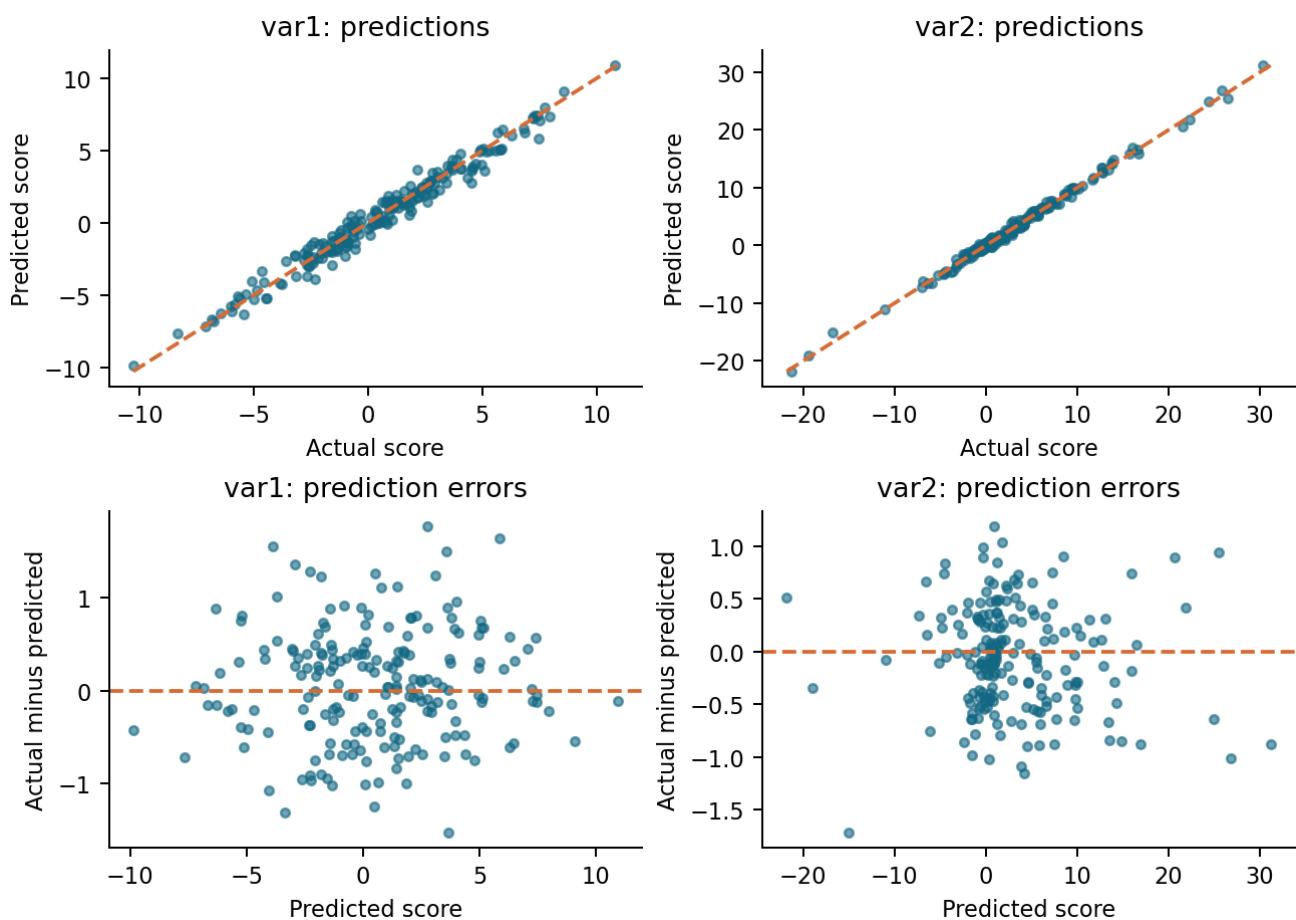


Figure 2. Top: predictions close to the diagonal line are accurate. Bottom: each point shows the prediction error. Errors close to zero are better.

5. What to keep in mind

Var1 test inputs are more often at the range boundaries than training inputs (98.2% versus 87.4% of rows have at least one input at -1 or 1). This difference may affect accuracy. The holdout is one reused split, so it is a useful check rather than a new independent evaluation.

Final files and code

Reproducing the work and preparing the submission

6. Final training and predictions

After choosing the settings, each model was fitted again using all 1,000 training rows. Predictions were saved in the original test-row order. Each submission contains one column named `y` and exactly 1,000 predictions. Predictions were not clipped or otherwise adjusted.

7. Reproduce the results

Use Python 3.12. Put the four dataset files and `sample_submission.csv` beside the scripts, then run:

```
python -m pip install -r requirements.txt
python train.py
python verify.py
python build_report.py
```

File	Purpose
<code>train.py</code> / <code>polynomial.py</code>	Train models and generate predictions
<code>compare_lasso.py</code>	Compare Lasso with Ridge on the same folds
<code>predict.py</code>	Make predictions from saved models
<code>verify.py</code>	Check calculations and prediction files
<code>outputs/metrics.json</code>	Save selected settings and results
<code>outputs/BT2024260_pred_var1.csv</code> <code>outputs/BT2024260_pred_var2.csv</code>	Prediction files for submission

8. Implementation and checks

NumPy and SciPy are used for Ridge, and scikit-learn is used for Lasso. Ridge minimizes $SSE + \alpha$ times the sum of squared coefficients. Lasso minimizes $SSE/(2n) + \alpha$ times the sum of absolute coefficients. SSE is the sum of squared errors and n is the number of fitting rows. The alpha values have different meanings.

Checks cover the Ridge calculations, the Lasso solution when selected, split separation, saved-model predictions, CSV columns, row counts, and row order. The original datasets are unchanged.

GitHub repository

https://github.com/Lohith600/ML_ASSIGNMNET